AI Operations Manager: Multi-Agent Pre-Sales & Business Workflow Automation System

Part 1 — AI Research & Evaluation

Problem Statement

The objective is to build an AI-powered business workflow automation system that can:

- Understand customer requirements
- Answer pre-sales queries
- Extract structured lead information
- Generate lead summaries
- Draft proposals
- Assist human sales teams

The core requirement is an LLM capable of:

- Natural conversation
- Information extraction
- Reasoning
- Structured JSON output
- Cost-efficient scaling

LLM Comparison

Criteria	OpenAI GPT-4o / GPT-5	Claude Sonnet	DeepSeek V4 Flash
Conversational Ability	Excellent	Excellent	Very Good
Structured JSON Extraction	Excellent	Excellent	Very Good
Business Reasoning	Excellent	Excellent	Good–Very Good
Speed	Fast	Fast	Very Fast
Context Handling	Excellent	Excellent	Good
Cost	High	Medium-High	Very Low
API Integration	Excellent	Excellent	Easy
Scalability	Excellent	Excellent	Excellent
Deployment Cost	Expensive	Moderate	Very Affordable

Criteria	OpenAI GPT-4o / GPT-5	Claude Sonnet	DeepSeek V4 Flash
Production Readiness	Excellent	Excellent	Good
Best Use Case	Premium enterprise applications	Research and analysis	Cost-sensitive automation systems

Tool 1 — OpenAI GPT-4o / GPT-5

Capabilities

- Advanced reasoning
- High-quality conversation
- Strong structured output generation
- Reliable function calling
- Excellent business understanding

Pricing

Higher than most alternatives.

Suitable when quality is more important than cost.

Scalability

Enterprise-grade.

Can support very large workloads.

Ease of Integration

Very easy.

Extensive documentation and SDK support.

Limitations

- Higher operational cost
- Can become expensive at scale

Best Use Cases

- Enterprise AI assistants
- Complex business workflows
- High-accuracy customer interactions

Tool 2 — Claude Sonnet

Capabilities

- Strong reasoning
- Excellent long-context handling
- High-quality business analysis

Pricing

Moderate to high.

Generally cheaper than premium OpenAI models.

Scalability

Suitable for enterprise deployments.

Ease of Integration

Simple API integration.

Limitations

- Higher cost than budget-focused models
- Not always necessary for simple workflow automation

Best Use Cases

- Research assistants
- Document analysis
- Long-form business reasoning

Tool 3 — DeepSeek V4 Flash (Selected)

Capabilities

- Fast response times
- Good reasoning ability
- Structured JSON generation
- Effective conversational performance
- Suitable for multi-agent workflows

Pricing

Very low compared to OpenAI and Claude.

Makes it suitable for high-volume automation systems.

Scalability

Can handle large numbers of requests while maintaining low operating costs.

Ease of Integration

Simple REST API integration.

Works well with automation platforms such as Make.com.

Limitations

- Slightly weaker reasoning than premium models
- Less consistent on highly complex business analysis tasks

Best Use Cases

- Workflow automation
- Customer engagement systems
- Lead qualification
- Cost-sensitive AI deployments

Tool Selection Decision

For this project, DeepSeek V4 Flash was selected as the primary model.

Reasons

1. Low operational cost

The system is designed for continuous customer interactions and high-volume usage. Low inference cost is important for scalability.

2. Fast response times

Real-time customer conversations require quick responses.

3. Strong structured output generation

The workflow relies heavily on JSON extraction for:

- Lead qualification
- Information extraction
- Agent-to-agent communication

4. Easy integration

DeepSeek integrates easily with Make.com through standard HTTP API calls.

5. Sufficient reasoning quality

Although OpenAI and Claude may provide stronger reasoning, DeepSeek offers the best balance between performance and cost for this use case.

Additional Workflow Platform Comparison

Criteria	Make.com	n8n	LangChain
Ease of Use	Excellent	Good	Moderate
Coding Required	Minimal	Low	High
Development Speed	Excellent	Good	Moderate
Multi-Agent Support	Good	Good	Excellent
Best For	Rapid business automation	Self-hosted automation	Advanced AI systems

Selected Platform: Make.com

Reasons:

- Fastest prototype development
- Native webhook support
- Easy API integrations
- Low development effort
- Suitable for demonstrating business workflow automation

Part 2 — Prototype / Proof of Concept

Project Title

AI Operations Manager: Multi-Agent Pre-Sales & Business Workflow Automation System

Objective

The objective of this prototype was to automate the pre-sales workflow of a software services company.

Traditionally, incoming customer inquiries require significant manual effort from sales teams to:

- Understand customer requirements
- Qualify leads
- Prepare summaries
- Draft proposals
- Route opportunities to the appropriate team member

The prototype automates these activities using AI agents and workflow automation.

Technologies Used

Workflow Orchestration

- Make.com

Communication Channel

- WhatsApp Cloud API

AI Model

- DeepSeek V4 Flash

Data Storage

- Make Data Store

Notification System

- Email

System Architecture

Customer sends a WhatsApp message.

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Webhook receives the message.

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Customer record is retrieved from Data Store.

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AI Qualification Agent processes the message.

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Lead information is extracted and stored.

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Once sufficient information is collected, the lead is marked as QUALIFIED.

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Lead Summary Agent generates a structured lead summary.

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Proposal Draft Agent generates a proposal outline.

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Sales team receives an email containing:

- Lead details
- Lead summary
- Proposal draft

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Customer is informed that the request is under review.

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Conversation state changes to WAITING_FOR_HUMAN.

AI Agents Implemented

Agent 1 — Lead Qualification Agent

Responsibilities:

- Answer customer questions
- Collect project requirements
- Extract structured information
- Qualify leads

Output:

- company_name
- service_required
- budget
- timeline
- requirements
- current_state

Agent 2 — Lead Summary Agent

Responsibilities:

- Analyze collected lead information
- Generate a concise business summary

Output:

- lead_summary

Agent 3 — Proposal Draft Agent

Responsibilities:

- Generate an initial proposal draft
- Suggest scope and next steps

Output:

- proposal_draft

Key Features

- WhatsApp-based customer interaction
- Persistent customer memory
- Automated lead qualification
- Structured information extraction
- Multi-agent workflow
- Automated proposal generation
- Human-in-the-loop review process
- Email notification system

Prototype Outcome

The system successfully demonstrates how AI agents can automate a significant portion of the pre-sales workflow while keeping a human reviewer in the final approval stage.

The prototype reduces manual effort, improves response times, and creates structured business information from unstructured customer conversations.

Part 3 — Recommendation Report

Recommended Architecture

The recommended architecture for the AI Operations Manager consists of the following components:

Customer (WhatsApp)

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WhatsApp Cloud API

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Make.com Workflow Engine

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Customer Memory Layer (Data Store)

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AI Qualification Agent

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Lead Summary Agent

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Proposal Draft Agent

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Email Notification System

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Human Sales Review

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Customer Follow-Up

Workflow

1. Customer initiates a conversation through WhatsApp.
2. Messages are received via WhatsApp Cloud API webhook.
3. Customer context and conversation history are retrieved from the Data Store.
4. The AI Qualification Agent answers customer queries and extracts structured lead information.
5. Once sufficient information is collected, the lead is marked as QUALIFIED.
6. The Lead Summary Agent generates a concise business summary.
7. The Proposal Draft Agent generates an initial proposal outline.
8. The sales team receives an email containing all generated outputs.
9. The customer is informed that the request is under review.
10. The lead enters WAITING_FOR_HUMAN state for manual follow-up.

Why Specific Tools Were Selected

DeepSeek V4 Flash

Selected because it provides:

- Low operational cost
- Fast response times
- Strong JSON generation capabilities
- Suitable performance for workflow automation
- Easy API integration

Although OpenAI and Claude offer stronger reasoning capabilities, DeepSeek provides the best balance between performance and cost for this use case.

Make.com

Selected because it provides:

- Rapid development
- Native webhook support
- Visual workflow orchestration
- Easy API integrations
- Minimal infrastructure requirements

This allowed the prototype to be built quickly while maintaining flexibility for future expansion.

WhatsApp Cloud API

Selected because:

- Customers are already comfortable using WhatsApp
- Real-world business applicability
- Direct integration with automation workflows
- Supports scalable customer communication

Make Data Store

Selected because:

- Simple implementation
- Persistent customer memory
- No additional database infrastructure required
- Suitable for prototype and small-scale deployments

Estimated Infrastructure Cost

Production Environment

Assuming:

- 1,000 leads per month
- Multiple AI agent executions per lead

Estimated monthly cost:

DeepSeek API:

- \$2-\$5/month ₹191.49 - ₹478.73/ month

Workflow Automation:

- \$53–\$91/month ₹5074.55 - ₹8712.90

Database:

- $0

Monitoring & Logging:

- $0

Estimated production cost:

- $60–$100/month

Actual costs depend on lead volume and model usage.

Risks and Limitations

AI Hallucinations

The AI may occasionally provide inaccurate or incomplete information if the required information is not available.

Mitigation:

- Human review before final proposal delivery.

Incomplete Customer Information

Some customers may not provide sufficient details for accurate qualification.

Mitigation:

- Follow-up questioning and human intervention.

Workflow Dependency

The system depends on external services including:

- WhatsApp Cloud API
- Make.com
- AI model provider

Mitigation:

- Retry mechanisms and monitoring.

Cost Growth

AI costs increase as conversation volume grows.

Mitigation:

- Use cost-efficient models such as DeepSeek.
- Reduce unnecessary AI calls.

Production Scalability

The architecture can be scaled by:

Database Upgrade

Replace Make Data Store with:

- PostgreSQL
- Supabase
- MySQL

for improved reliability and analytics.

Dedicated Backend

Introduce:

- FastAPI
- Node.js

to manage workflows and business logic.

Multi-Agent Expansion

Additional agents can be added for:

- Risk analysis
- Solution architecture
- Project estimation
- Contract drafting

Knowledge Base Integration

Add a RAG layer using:

- Pinecone
- Weaviate

to provide company-specific answers.

CRM Integration

Integrate with:

- HubSpot
- Zoho CRM
- Salesforce

for automated lead management.

Monitoring and Observability

Implement:

- Logging
- Analytics dashboards
- Error monitoring

for production reliability.

Conclusion

The proposed AI Operations Manager successfully demonstrates how AI agents can automate significant portions of the pre-sales and business operations workflow. The architecture is cost-effective, scalable, and designed around a human-in-the-loop approach, ensuring both efficiency and operational control.